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Safe Backup Control Synthesis for a Multicopter System

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Why Safety Matters

- Autonomous systems operate in environments where failures may cause damage to humans and infrastructure.
- For multirotors...
 - Fast and agile dynamics
 - Cluttered environment
 - Limited thrust
 - Sensitive to disturbances

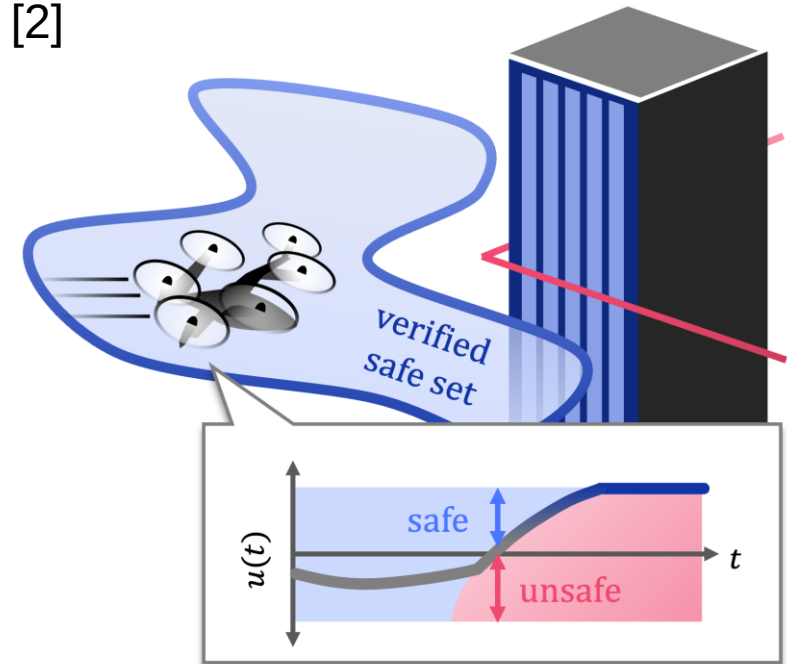
[1]



What is Safety in Control?

- **Safety:** The system state remains within an admissible set at all times.
- **Safe set $\mathcal{X}_s$:** Set of states satisfying safety constraints.
- However, being inside the safe set now does not necessarily guarantee that the state will remain safe in the future.
- **Safety condition:** $x(0) \in \mathcal{X}_s \Rightarrow x(t) \in \mathcal{X}_s, \quad \forall t \geq 0.$
i.e., a trajectory starting in $\mathcal{X}_s$ never leaves $\mathcal{X}_s$.
 $\rightarrow \mathcal{X}_s$ is (forward) invariant.

[2]

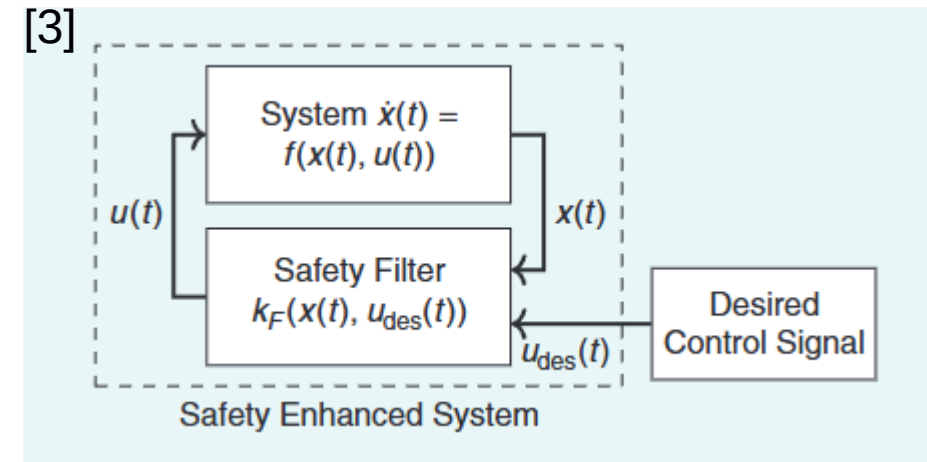


Existing Approaches for Safety

- **Hamilton–Jacobi Reachability:** Rigorous safety analysis
Limitation: Curse of dimensionality
- **Constrained Optimal Control:** Directly enforces state and input constraints
Limitation: High online computational burden
- **Safety Filtering:** Separates safety enforcement from the nominal controller

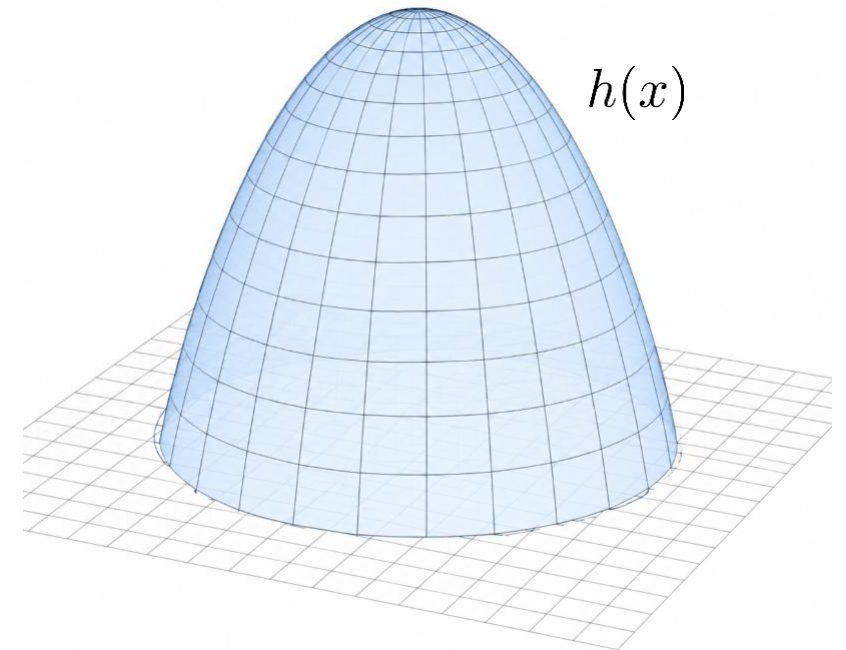
Advantages:

- Independent of the nominal controller
- Safety constraints enforced only when necessary
- Can incorporate different safety verification methods



Control Barrier Function (CBF)

- **Control Barrier Function (CBF):** A function that characterizes the boundary of a safe set. → widely used for safety filtering
- **Safe set:** $\mathcal{X}_s = \{x : h(x) \geq 0\}$
- **Safety boundary:** $h(x) = 0$
- **Safety condition:** $\dot{h}(x, u) + \alpha(h(x)) \geq 0$ **(1)**
- If the safety condition is satisfied, the system is prevented from leaving the safe set.



CBF-QP as a Safety Filter

- **Nominal input** u_{nom} : Control input generated by the original controller

- **Quadratic Program (QP):**

$$\begin{aligned} u_{\text{safe}} &= \arg \min_{u \in \mathcal{U}} \|u - u_{\text{nom}}\|^2 \\ \text{s.t.} \quad &\dot{h}(x, u) + \alpha(h(x)) \geq 0 \end{aligned}$$

- CBF-QP projects the nominal control input onto the set of safe admissible control inputs.
→ minimal intervention
- **Challenge:** Constructing a valid CBF $h(x)$ can be difficult for systems with complex dynamics and input constraints.

$$u_{\text{nom}} \rightarrow \boxed{\text{CBF-QP Safety Filter}} \rightarrow u_{\text{safe}}$$

Backup-Based Safety Certification

- **Why backup?** Instead of constructing a safety certificate directly, check whether a feasible recovery maneuver can safely reach a region where safety can be maintained indefinitely.
- **Safety constraint:** Suppose the safety requirement is represented by $\ell(x)$.
- **Backup policy** κ_b : Admissible policy used to recover toward a known safe region
- **Backup rollout** $\phi_b(t, x)$: Predicted trajectory under the backup policy
- **Safety certification:** The rollout remains safe and reaches backup set within the backup horizon
- **Backup-based CBF:**
$$h_I(x) = \min \left\{ \underbrace{\min_{0 \leq t \leq T_b} \ell(\phi_b(t, x))}_{\text{Safe during recovery}}, \underbrace{h_b(\phi_b(T_b, x))}_{\text{Reach the backup set}} \right\}$$
- **Challenge:** Backup policy design, conservatism of fixed backup policy.

Proposed: Time-Optimal Backup Control

- **Time-optimal backup:** Generates the fastest feasible recovery trajectory to a safe invariant set

- Solve optimization problem:

$$\begin{aligned} & \min_{u(\cdot), T} T \\ \text{subject to} \quad & T \geq 0, \quad x(0) = x_0 \\ & \dot{x}(t) = f(x(t)) + g(x(t))u(t) \\ & u(t) \in \mathcal{U}, \quad x(t) \in \mathcal{X}, \quad \forall t \in [0, T] \\ & x(T) \in \mathcal{X}_s \end{aligned}$$

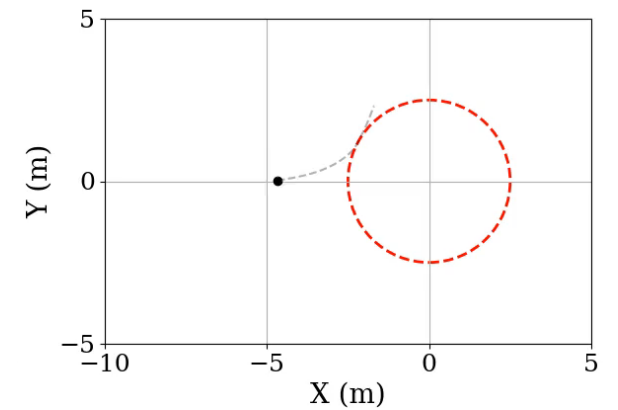
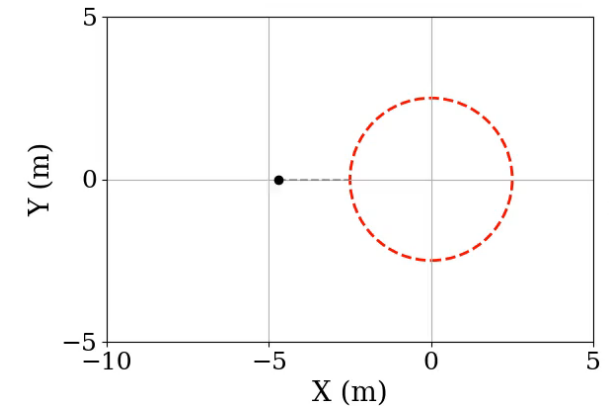
- The solution provides a backup trajectory that guarantees minimum-time recovery.
- The backup maneuver is synthesized online for the current situation.

Algorithm

- **Online backup synthesis:** Solve the time-optimal control problem from the current state
- **Adaptive backup horizon:** Obtain the optimal recovery time $T^*(x)$
- **Backup rollout:** Generate the optimal trajectory $\phi^*(t, x)$ toward the safe invariant set
- **Construct CBF:**
$$h_I(x) = \min\left\{\min_{0 \leq t \leq T^*(x)} \ell(\phi^*(t, x)), h_b(\phi^*(T^*(x), x))\right\}$$
- **Safety filtering:** Use the backup-based barrier function in the CBF-QP

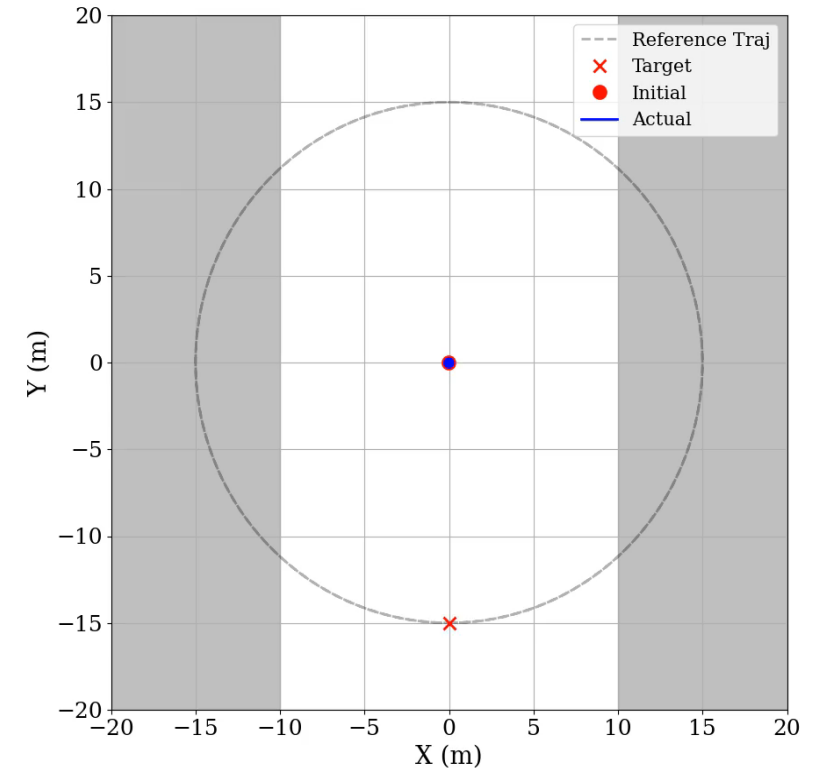
Simulation

- **Setup:** 2 kg multirotor with bounded thrust 0–40N and body angular rates $\pi[0.5,0.5,0.3]$ rad/s
- **Backup Trajectory Comparison**
 - **Scenario:** Same initial position near a cylindrical obstacle, but different initial velocities
 - **Upper:** Initial velocity $[5.0,0.0,0.0]$ m/s
Result: Maximum braking $\rightarrow$ stops just before the obstacle boundary
 - **Lower:** Initial velocity $[6.0,1.0,0.0]$ m/s
Result: Braking alone is insufficient $\rightarrow$ steers around the obstacle and stops safely
 - Time-optimal control generates a situation-dependent backup trajectory.



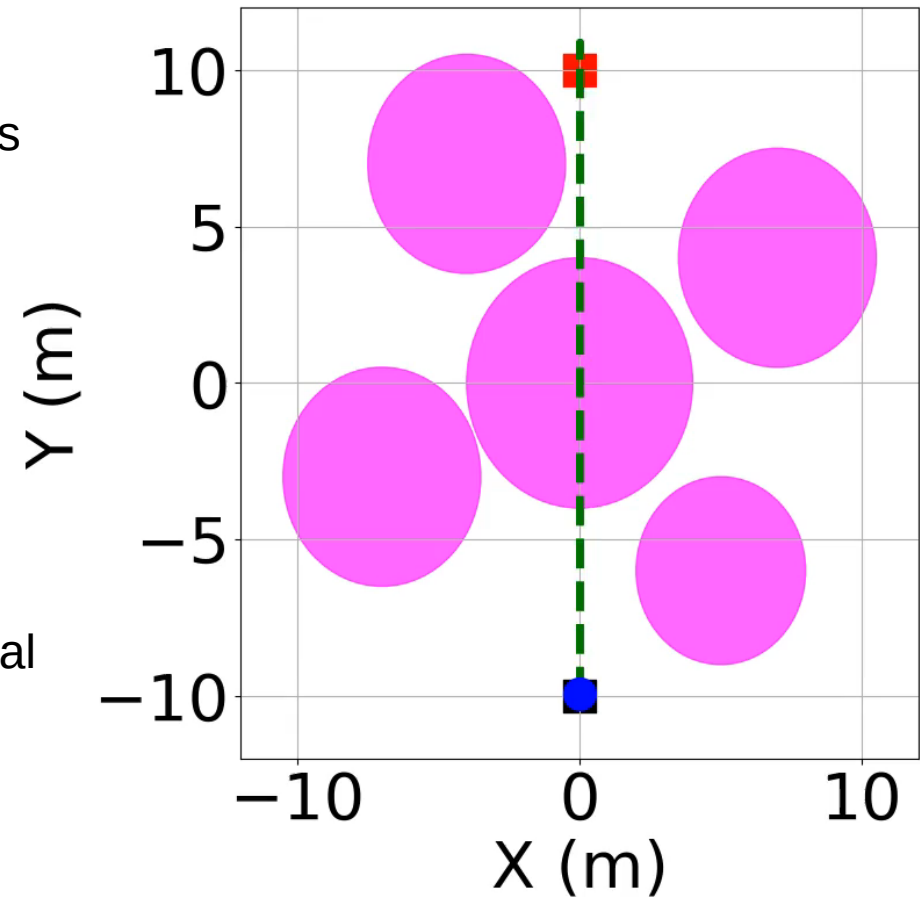
Simulation

- **Safety Filtering Result**
 - **Scenario:** Track a circular reference trajectory with a radius of 15m, intentionally extending beyond the safe region
 - **Switching filter:** Apply the time-optimal backup input when the minimum constraint value along the backup rollout falls below 0.2
 - **Nominal trajectory:** Violates the safety constraint
 - **Filtered trajectory:** Remains inside the safe region and moves along the constraint boundary
 - **Result:** The proposed backup utilizes the available safe region without unnecessary stopping



Simulation

- **Obstacle avoidance**
 - **Nominal controller:** Nonlinear MPC without safety constraints
 - **Nominal behavior:** Attempts to penetrate obstacles while moving toward the goal
 - **Filtered behavior:** Avoids obstacles while satisfying the constraints
 - **Computation time:** Average TOC solve time of 23.76ms
 - **Result:** Safety is maintained despite potentially unsafe nominal commands



Conclusion

- We propose an online time-optimal backup control framework for multirotor safety.
- The recovery trajectory and horizon adapt to the current state.
- The backup trajectory is incorporated into a safety filter to satisfy state and input constraints.
- Simulation results demonstrate safe maneuver of a multirotor.
- Future work includes backup set design strategy and hardware validation.

Thank you!

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